

Wazuh SOC Lab Setup Documentation

Architecture Overview

Kali Linux (Agent) → Wazuh Manager → Wazuh Indexer → Wazuh Dashboard

1. Requirements

- Laptop or PC with at least 8 GB RAM
- Oracle VirtualBox installed
- Kali Linux Virtual Machine
- Wazuh OVA image

2. Downloading Wazuh OVA

Download the Wazuh OVA file from the official documentation website. This OVA contains the Wazuh Manager, Indexer, and Dashboard preconfigured.

Downloaded : wazuh-4.14.3 , later imported to virtual box

Steps:

- 1 Open Oracle VirtualBox.
- 2 Click File → Import Appliance.
- 3 Select the downloaded Wazuh OVA file.
- 4 Click Next → Import.
- 5 Start the Wazuh virtual machine.

3. Accessing the Wazuh Server

Login credentials used in the server terminal:

```
Username: wazuh-user
```

```
Password: wazuh
```

Check the server IP address:

```
ip a
```

4. Accessing the Wazuh Dashboard

Open a browser on the host machine and navigate to the Wazuh dashboard using the server IP address.

In Browser : `http:// (ip add)`

Default dashboard credentials:

Username: admin

Password: admin

5. Deploying the Wazuh Agent on Kali Linux

Agent installation command used:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.3-1_amd64.deb && sudo WAZUH_MANAGER='10.156.213.239' WAZUH_AGENT_NAME='kali-agent' dpkg -i ./wazuh-agent_4.14.3-1_amd64.deb
```

Enable and start the agent service:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable wazuh-agent
```

```
sudo systemctl start wazuh-agent
```

6. Verifying Agent Connection

After installing the agent, open the Wazuh dashboard and navigate to:

Endpoints → Agents

The Kali agent should appear with status 'Active'.

7. Generating Security Alerts (Test)

To test detection, failed SSH login attempts were generated.

```
ssh fakeuser@localhost
```

After multiple failed login attempts, Wazuh generated authentication failure alerts in the dashboard.

8. Conclusion

The SOC lab was successfully created using Wazuh and Kali Linux. The Wazuh agent sends logs to the manager where they are analyzed and visualized in the dashboard. This setup demonstrates basic security monitoring and attack detection capabilities.

Downloaded :

OS : linux : DEB amd64

Server Add :

Agent name : kali-agent





